Adversarial Feature Equilibrium Network for Multimodal Change Detection in Heterogeneous Remote Sensing Images

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***Abstract*— Change detection (CD) methods have been crucial in exploring geo-environmental science. With the advancement of remote sensing (RS) technology, multimodal images acquired from different platforms and sensors are widely used for CD tasks. As an emerging task, multimodal CD (MCD) aims to achieve more comprehensive and precise detection of land cover changes through complementary information in multimodal images. However, there are significant differences between modalities, particularly in heterogeneous images. How to deal with modal differences while effectively integrating change information remains a challenge in MCD. In this article, we propose a novel adversarial feature equilibrium network (AFENet), which establishes an additional adversarial optimization to solve the equilibrium problem between modal differences and land cover changes. Our AFENet aligns the features and reduces the modal gap through a multiscale adversarial domain adaptation (MADA) approach. Meanwhile, a divergence-aware contrastive module (DCM) is designed as a regularization term for adversarial optimization. DCM affects the sensitivity of feature extractors by constraining the mutual information between changed and unchanged pixels. In this case, AFENet can maintain the consistency of feature representation while maximizing the discriminability of change targets. The features extracted from AFENet will then be integrated by our multistream feature fusion (MFF) module and utilized to generate change maps. The effectiveness of our approach is demonstrated on two scene-level multimodal RS datasets. Compared with existing methods, our AFENet achieves state-of-the-art (SOTA) performance on both datasets and outperforms the second-best $F1$ score by 4.64% and 1.1%, respectively.**

***Index Terms*— Change detection (CD), domain adaptation (DA), heterogeneous image, multimodal, remote sensing (RS).**

NOMENCLATURE

Abbreviations	
RS	Remote sensing.
CD	Change detection.
MCD	Multimodal change detection.
AFENet	Adversarial feature equilibrium network.
MADA	Multiscale adversarial domain adaptation.
DCM	Divergence-aware contrastive module.
DL	Deep learning.
MFF	Multistream feature fusion.
CAF	Cross-attention fusion.
AFE	Adversarial feature equilibrium.
GADA	Generative adversarial domain adaptation.
ADA	Adversarial domain adaptation.
Symbols	
X_1,X_2	Bi-temporal images.
Y	Change mask.
$\mathbb{D}_1$,$\mathbb{D}_2$	Data domain of X_1,X_2.
G_1,G_2	Feature encoder networks of X_1,X_2.
D_{dis}	Domain discriminator.
D_{cd}	Decoder network for change detection.
f_k^l	Feature maps of k domain at lth scale.
P	Predicted probabilities.
h_k^l	Joint feature of f_k^l and P.

I. INTRODUCTION

REMOTE SENSING (RS) technology utilizes aviation or satellites to collect environmental electromagnetic wave information, enabling the monitoring and identifying environmental quality conditions for distant surface targets. This advanced environmental information acquisition technology has led to the growth of Earth observation programs such as Sentinel, WorldView, and GaoFen [1], [2], [3]. These programs provide a vast amount of RS images with varying resolutions

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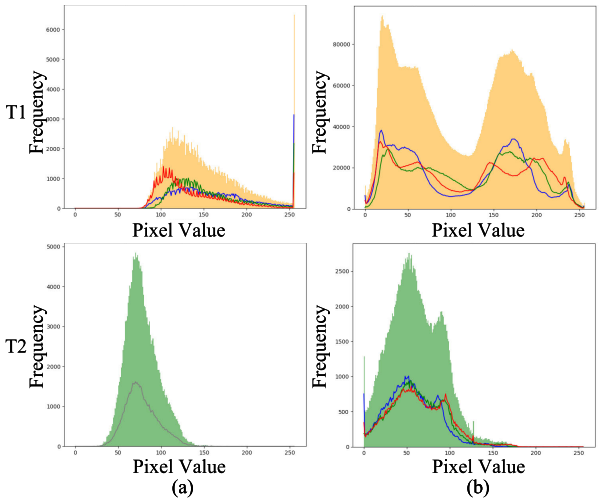


Fig. 1. Statistical histograms of the bi-temporal images in multimodal datasets. (a) Feature distribution of the MT-Wuhan dataset and (b) feature distribution of the MT-HTCD dataset. The details of the two datasets will be introduced in Section IV. Rows T1 and T2 represent the pre-change images and the post-change images. There are significant distribution differences between bi-temporal images, which reflects the modal differences in multimodal data. Our motivation is to address the impact of the modal difference problem on MCD.

and modalities. By utilizing these data to explore the changes occurring on the surface, we can gain a deeper insight into the Earth's ecology, human activities, and so on. RS CD technology [4] provides essential support in environmental detection, disaster warning, and land planning.

With the advancement of RS technology, multimodal images acquired from different platforms and sensors have gradually begun to be applied to CD tasks. Standard CD tasks mainly rely on single-modal data, such as optical and radar images. Limited by the single data source, standard CD may struggle to perceive the complete land cover types and capture the surface changes comprehensively. For instance, bi-temporal images gathered by optical sensors have very high resolution but can be affected by weather, lighting conditions, and other factors, which will degrade the detection performance. In contrast, SAR images have the ability to penetrate clouds, operate at night, and provide structural information but tend to have high image noise, which can also affect the precision of CD. Besides, the high cost and low frequency of RS data collection may lead to a lack of the same modality images at different observation epochs. As an emerging task, MCD aims to address the limitations of standard CD by utilizing bi-temporal images of different modalities. The advantage of the MCD task is that, by integrating complementary information from multimodal images, it can achieve more comprehensive and accurate detection of land cover changes as well as adapt to complex detection scenarios.

In recent years, the superiority of DL methods over traditional ones has been widely validated in CD tasks, especially when dealing with large-scale scene-level data. Because scene-level RS data usually contains complex land cover types and environmental conditions, which pose higher requirements on the feature representation capabilities of detection methods. Most traditional methods rely on manually designed features that make it difficult to capture complex change patterns and traditional methods are linear or nearly linear computational processes, which leads to weak nonlinear modeling capability [5], [6], [7], [8]. In contrast, DL methods can learn complex nonlinear feature representations through multilayer structures, thus effectively capturing change information. Therefore, an increasing number of researchers have proposed DL-based approaches to address the difficulties in the MCD task [9], [10], [11], [12], [13], [14]. For example, Lv et al. [15] introduce a simple UNet-based CD network, which mainly contains a multiscale convolution module and a combined loss function, to solve the heterogeneous image problem. Jiang et al. [16] propose a novel homogeneous transformation model based on image style transfer (IST) called deep homogeneous feature fusion (DHFF). DHFF separates semantic content and style features in heterogeneous images during homogeneous transformation to prevent the corruption of semantic information. Ebel et al. [17] design a novel architecture that incorporates both optical and SAR data and utilizes a UNet-based Siamese network to process the multimodal information. This article aims to improve the accuracy of MCD by supervised DL methods. Despite advancements, current methods still encounter the following challenges.

1) The RS images of different modals exhibit significant differences due to variations in acquisition sensors and imaging mechanisms, which leads to variations in observation characteristics and statistical distributions between images, as shown in Fig. 1. Especially in heterogeneous situations, excessive modality differences can interfere with the network learning direction and degrade detection performance. Thus, how to address modality differences is a crucial problem for MCD.
2) In order to leverage the benefits of MCD, the network needs to preserve the complementary information between different modalities, particularly the information related to change targets. However, a natural equilibrium problem exists between domain-invariant features and geophysical change features during feature extraction. Retaining and integrating geophysical change features while reducing the modality gap are critical challenges for improving detection accuracy.

In this article, the AFENet is proposed to address the aforementioned challenges. By treating the bi-temporal multimodal images as different domains, our MADA is utilized to align features and reduce the modal difference. Meanwhile, a DCM is designed to constrain the mutual information between changed and unchanged pixels, which can adjust the sensitivity of the feature extractor. In this case, AFENet can maintain the consistency of feature representation while maximizing the discriminability of change targets. Then, the multimodal feature is integrated by our MFF module and utilized to generate change maps.

The main contributions of this article are summarized as follows.

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1) Focusing on the MCD task, we propose an end-to-end CD framework called AFENet, which establishes an additional adversarial optimization strategy to solve the equilibrium problem between modal differences and land cover changes. AFENet aligns the features and reduces modal differences by MADA. Meanwhile, the land cover changes are enhanced by our designed DCM during the adversarial process. In this case, AFENet can maintain the consistency of feature representation while maximizing the discriminability of change targets.
2) Our DCM is designed as a regularization term for adversarial optimization. In this work, we further adapt the supervised contrastive learning (SCL) to pixel level and adjust feature differences by Euclidean distance with threshold. DCM constructs feature pairs of changed and unchanged pixels and dynamically adjusts the feature differences, which in turn affects the sensitivity of the feature extraction network to the change information.
3) The multistream features extracted by AFENet are integrated through our MFF and used to generate change maps. In MFF operation, we introduce a CAF strategy to capture the correlation and enhance the features effectively. The feature interaction helps the model to understand the complex patterns and structures in the data, thereby improving the prediction accuracy. Additionally, we conduct a comparative analysis of the CAF strategy against three other fusion strategies to verify the effectiveness of CAF in the MCD task.

The rest of this article is organized as follows. Section II presents the existing works related to the MCD task. In Section III, we introduce the overall structure of AFENet and the technical details of each module. Section IV compares AFENet with existing classical methods and verifies its effectiveness through ablation experiments. In Section V, we summarize the contributions of this article and outline future research work.

II. Related Work

In this section, we will review the several techniques used in this article and discuss their applications to CD.

A. Supervised DL-Based CD

DL methods have demonstrated exceptional performance in RS tasks, such as scene classification, object detection, and segmentation. Among the DL methods, supervised models tend to achieve highly accurate performance with the support of labeled data and computing resources. Therefore, many supervised approaches have been employed in the CD task [18], [19], [20], [21], [22]. In terms of network structure, these approaches can be categorized into Siamese and non-Siamese networks. The Siamese network processes the bi-temporal images through two weight-sharing sub-networks, which ensures symmetry and consistency in the feature extraction process. This approach can capture the local changes effectively and shows good performance in scenes with relatively simple change patterns. For example, Daudt et al. [23] present three end-to-end supervised frameworks based on the fully convolutional network and use skip connection in two Siamese networks for the first time. Chen et al. [24] propose a deep Siamese convolutional multilayer recurrent neural network (SiamCRNN), which combines the advantages of Siamese networks and recurrent neural networks to extract the temporal features. Fang et al. [25] propose a densely connected Siamese network based on the NestedUNet [26] to better retain the high-resolution and fine-grained representations. As for the non-Siamese network, it processes the input images through two independent branches or one unified network. The independent branches can flexibly design the two feature extractors, which can extract the detail change information while better adapting to the bi-temporal images with significant differences. Zheng et al. [27] propose a non-Siamese network with two branches to improve the accuracy and robustness of the CD by transferring spectral information between the two branches through spectral knowledge transfer (SKT). Besides, the unified network shares the same feature extractor, which is computationally efficient and global information sensitive. Such as Peng et al. [28] propose a difference-enhancement dense-attention convolutional neural network (DDCNN), which designs the feature extractor based on the UNet++ structure. The bi-temporal image transformer (BIT) method uses a unified ResNet as the backbone to extract high-level semantic features [29]. Wang et al. [30] present a general end-to-end 2-D convolutional neural network (GETNET) to extract the discriminative features and handle the high-dimension problem for hyperspectral images.

The above-mentioned supervised CD methods and the MCD methods discussed in Section I both extract features through a single extraction network structure, followed by enhancing the feature representation through additional modules. In order to address the challenges in MCD, we integrated independent branches and a unified network to design a multistream structure within our AFENet. As shown in the blue block in Fig. 2, the bi-temporal images are input into two independent feature extraction networks, while the concatenated image is inputted into a unified network. This network design is able to capture local and global change information more comprehensively within the feature space and integrates this information through effective feature fusion strategies.

B. Feature Fusion in CD

Feature fusion is a common operation of information integration, which is extensively applied in DL CD methods. Based on the fusion position, the existing methods can be categorized into early fusion, middle fusion, and late fusion [31]. Middle fusion refers to fusing the middle layer features of the network, which combines complementary information from layers to improve the discriminative power of features and enhance the nonlinear modeling capabilities of the model. Two critical considerations in the design of middle fusion are the structure of the feature extraction network and the fusion strategy. For network structure, there have been several works validating the effectiveness of ResNet [32], ViT [33],

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UNet [34], etc. in the CD task. Thanks to the symmetric structure of the encoder–decoder and the design of skip connection, UNet has excellent multiscale feature extraction capability and flexible feature fusion capability. Therefore, the unified network in our AFENet follows the structural settings of UNet, with modifications made only to the number of feature channels. The two independent branches have the same structure as the encoder part of the unified network, but the weights are not shared. Such a multistream feature extraction approach can construct richer feature representation. In addition, we design an ADA approach for extracting and aligning features for two branches of AFENet. By confusing a domain discriminator, the bi-temporal images are brought closer in the feature space, thereby mitigating the effects of modal differences.

As for the feature fusion strategy, the common ways are feature concatenation, feature difference, and weighted average, for example, [35], [36], and [37]. In addition, there are fusion strategies based on attention mechanisms that aim to enhance important information during the fusion process [38]. For example, squeeze-and-excitation networks (SENets) selectively improve the representation of important feature channels by generating channel-level attention weights [39]. The convolutional block attention module (CBAM) enhances spatial location information in feature maps by combining the channel attention module and spatial attention module [40]. Hou et al. [41] propose a novel coordinate attention mechanism, which computes the attention weights in both the length and width directions of the feature map separately, preserving the positional information of the features while capturing the spatial and channel information. Huang et al. [42] propose a criss-cross attention mechanism that captures contextual information over long distances by calculating the spatial relationship between each pixel and all the pixels in its cross position, which improves the global feature representation. Inspired by existing works, we introduce a CAF strategy, which applies the channel weights learned from the unified network to the features of two independent branches, and then fuses the features from triple streams. Through this feature interaction, the semantic information of the change targets in the feature maps is enhanced. We validate the effectiveness of CAF by comparing it with three other commonly used fusion strategies in MCD tasks.

C. ADA for MCD

Domain adaptation (DA) is a crucial aspect of transfer learning, aiming to solve the problem of distributional differences between source and target domains. This allows the model trained on the source domain data to be directly applied to the target domain data. However, measuring the distance between the two domains and reducing the discrepancy between them are the main challenges in DA. Existing DA methods mainly contain alignment-based methods, reconstruction-based methods, and adversarial learning-based methods [43], [44], [45], [46], [47]. Among them, the methods based on adversarial learning can be subdivided into two categories: GADA and ADA. GADA generates data in the style of the target domain through GAN or directly aligns the data distribution between the source domain and the target domain at the pixel level to improve the generalization ability of the model. For example, Hoffman et al. [48] proposed a cycle-consistent ADA (CyCADA), which achieves effective adaptation at the pixel level and feature level by combining a GADA network with a cycle consistency module. Hu et al. [49] generate domain-invariant representation by a GAN architecture with duplex adversarial discriminators. This method achieves pixel-level alignment by simultaneously transforming the images in the source and target domains to each other with two generators and two discriminators. As for ADA, it mainly focuses on feature-level alignment. Tzeng et al. [45] first proposed a generalized ADA framework called adversarial discriminative DA (ADDA). ADDA guides the learning of the feature extractor by confusing the domain discriminator, aligning the feature distribution and thus improving the task performance in the target domain. Based on the above work, Long et al. [50] introduce the discriminative information in classifier prediction into ADA and propose conditional domain adversarial networks (CDANs). CDAN combines adversarial training and conditional information to effectively align the conditional distributions of the source and target domains. Pei et al. [51] propose a multiADA method to finely align the category-conditional distributions. This approach captures the feature distribution of different categories through multiple adversarial discriminators, where each discriminator focuses on a specific category. Xu et al. [52] introduce the data augmentation strategy into DA and propose an ADA with domain mixup (DM-ADA). During the adversarial training process, this approach performs domain mixup at the pixel level and feature level by linear interpolation, which improves the adaptability of the model.

In the context of MCD, the significant factor contributing to poor detection performance is the distributional difference between bi-temporal images and DA methods have the ability to handle significant differences and align features between source and target domains. Therefore, it is reasonable to consider employing DA approaches as the solution to address the modal difference in MCD. Some researchers have published related works. Chen et al. [53] propose a cross-domain CD method called DADANet, which extracts spatial–spectral features through a Siamese convolutional network and matches the feature representations of the two domains through multikernel maximum mean discrepancy (MK-MMD). MK-MMD is an extension of MMD, which uses multiple kernel functions to compare the mean embeddings of two distributions in the reproducing kernel Hilbert space (RKHS) to determine their similarity. Zhang et al. [54] apply the MK-MMD-based DA to heterogeneous CD. This work extracts features through two branches that do not share weights and constrains the features using MK-MMD-based DA, effectively reducing the domain differences between heterogeneous data. Liu et al. [55] propose a supervised end-to-end CD method that achieves image-level adaptation through GADA based on cycle consistency, thereby effectively achieving training sample augmentation. Niu et al. [56] used a conditional GAN

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to transform heterogeneous images into more consistent image representations. The generated image pairs can be directly compared as homogeneous images.

Section II-B explains that the performance of MCD is greatly affected by the quality of features. Therefore, in order to effectively extract and fuse multimodal data features, we align the features through the ADA framework. Inspired by existing works, we design our MADA based on CDAN [50]. The introduction of conditional information is to better retain the discriminative information of the change category during the feature alignment process, making the alignment process more targeted. The multiscale strategy is to capture local and global information more comprehensively while improving the stability of adversarial training to adapt to different data domains. The feature maps extracted by MADA are then integrated through the CAF fusion strategy, which can effectively improve the accuracy and robustness of our AFENet.

D. Contrastive Learning

Contrastive learning, as a learning paradigm, is widely used in computer vision tasks such as image classification and segmentation. This paradigm learns the feature representation of data by constructing positive and negative sample pairs. Traditional contrastive learning methods learn robust and general feature representations in an unsupervised or self-supervised approach, thereby improving the performance and generalization ability of the model in visual tasks. For instance, Chen et al. [57] propose the SimCLR method, which augments images from different views and learns visual representations by directly comparing the features of image pairs within a single batch. The momentum contrast (MoCo) method proposed by He et al. [58] greatly improves the performance of unsupervised learning in vision tasks by constructing a dynamic dictionary with a queue and a moving-averaged encoder. Many self-SCL methods are introduced in [59]. Khosla et al. [60] introduce category labels as supervisory information into the original CL method and SCL. This approach shortens the feature distance of the same categories and increases the feature distance of different categories to learn more discriminative features, thereby improving the classification performance of the model.

The works in [61] have verified the influence of regularization terms on adversarial learning. During adversarial training, the regularization term guides the direction of the learning by affecting the parameter update. Appropriate regularization terms can help the model learn the target features effectively. Based on this idea, we propose DCM as the regularization term of MADA, which is an extension of SCL. To better adapt to the MCD task, our DCM extends SCL to pixel-level granularity and constrains feature differences by Euclidean distance with thresholding. Our goal is to improve the perception of MADA to the change information, thereby enhancing the discriminative ability of features. Finally, DCM and MADA constitute our AFENet, which solves the equilibrium problem between modal differences and land cover change differences, allowing the feature extractor to generate more discriminative alignment features. The features are then integrated through the CAF strategy, which effectively improves the accuracy and robustness of the MCD model.

III. Methodology

In this section, we will introduce the overall structure of the proposed MCD method and describe each component in detail.

A. Overall Architecture

The overall architecture of our AFENet is shown in Fig. 2 (a), which can be mainly divided into two parts: the AFE in Fig. 2(b) and the MFF for MCD in Fig. 2(c). The CD network in part (c) consists of a unified network and two independent branches. Since the weights are not shared among the three networks, AFENet can be categorized as a non-Siamese CD framework. The unified network is built on a five-layer UNet architecture, which realizes feature extraction and change map generation through an encoder and a CD decoder network. The structure of two branches is the same as the encoder part of the unified network, aiming to extract multiscale features for use in AFE and MFF. The adversarial network in part (b) is based on a conditional adversarial architecture, which consists of conditional operations, discriminators, and the DCM. The details of each component can be seen in the color blocks.

Bi-temporal images are input into three independent encoder networks to extract semantic features at different levels. AFE affects the perception of two encoder networks by the collective effect of MADA and the DCM. By confusing the domain discriminator, the feature encoder networks are forced to align the domain-invariant features and retain the discriminability of the change target. Meanwhile, the extracted feature maps are integrated by the CAF strategy in the MFF. The fused feature is fed into the CD decoder network to generate the change map.

B. Adversarial Feature Equilibrium

For multimodal images, there are modal differences and land cover changes between change regions. The modal differences are mainly caused by different acquisition sensors and imaging mechanisms, which can directly affect the accuracy of the CD model. Even with supervised learning, the network struggles to recognize the change features from modal differences. In DA tasks, it is a commonly used approach to reduce the domain gap by aligning and matching features in Hilbert space. Inspired by the existing works, we treat multimodal images as separate data domains and aim to reduce the impact of modal differences through the ADA approach. However, it is difficult for the ADA framework to align the distributions of two domains stably, even if the domain discriminator has been confused. This instability issue leads to the absence of land cover change features, which can limit the performance of the detection model. Our AFE is proposed to address the equilibrium problem between modal difference and land cover

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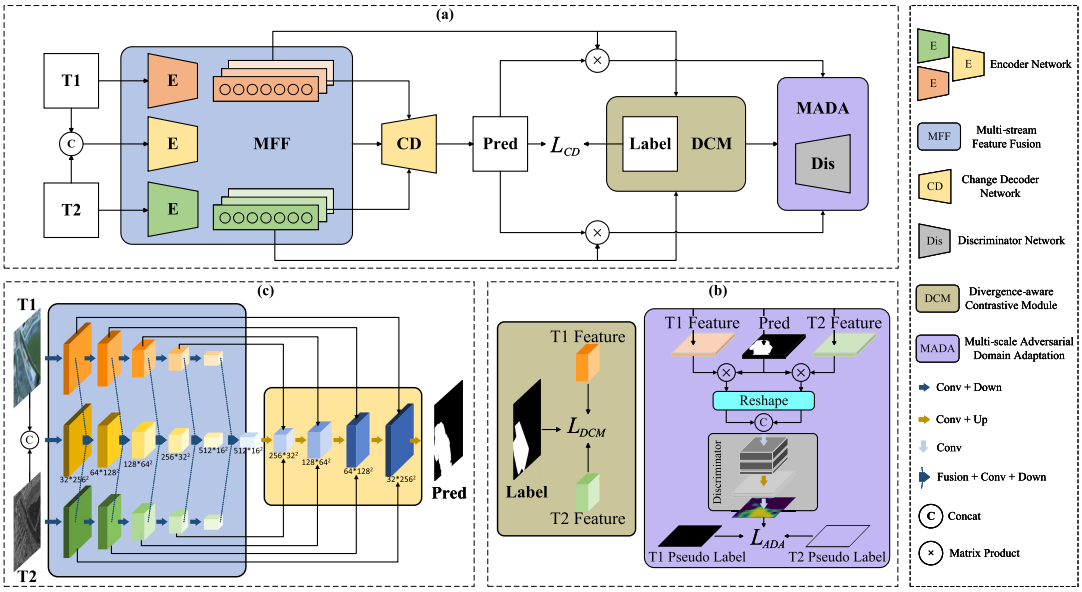


Fig. 2. Overall architecture of the proposed AFENet. The overall architecture and components of the proposed AFENet. (a) Framework and operation flow of AFENet. (b) AFENet mainly consists of the AFE and (c) MFF for MCD. The AFE is composed of the MADA and its additional regularization term DCM. The extracted features from three independent encoder networks are integrated through the MFF and the fused feature is used to generate change maps through the decoder network. The details of each component are shown in the corresponding color blocks.

change by establishing an additional adversarial optimization through MADA and DCM.

The MADA is an extension of the CDAN, which consists of discriminators corresponding to the number of feature scales. Each discriminator is a three-layer network structure consisting of deconvolutional layers and up-sampling operations. The network output is then transformed into domain prediction probabilities by a sigmoid function. The final output of the discriminator is the loss of binary cross-entropy (BCE). Meanwhile, DCM is designed as the regularization term of MADA, which aims to enhance the discriminability of features through a pixel-level contrastive constraint loss function. Through the adversarial effect between MADA and DCM, the network will eventually reach an equilibrium state, which can maintain the consistency of feature representation while maximizing the discriminability of change targets.

1) Multiscale ADA: In the context of bi-temporal RS images with different modalities, we consider them as two data domains, $X_1 \in \mathbb{D}_1$ and $X_2 \in \mathbb{D}_2$. Features for adversarial learning are extracted by feature encoder networks G_1 and G_2, respectively. Besides, the domain discriminator D_{dis} is used to determine whether the feature $f_k^l = \{G_k(x_k)|k = 1, 2\}$ is from the data domain $\mathbb{D}_1$ or $\mathbb{D}_2$. The training process can be formalized as an adversarial optimization problem in which the feature extractor and the discriminator jointly optimize an objective function, which usually consists of two parts

$$\mathcal{L}_{\text{ADA}} = \mathcal{L}_{\text{CD}} - \lambda \cdot \mathcal{L}_{\text{DA}} \tag{1}$$

where L_{CD} represents the CD task loss, L_{DA} represents the domain adversarial loss, and λ is the adversarial weight coefficient. By jointly optimizing the task loss and domain adversarial loss, the feature extractor is forced to learn a domain-invariant, task-beneficial feature representation between the two domains. Guided by the conditional generative adversarial network (CGAN) and CDAN, we introduce additional conditional information in the adversarial network with the aim of making the adversarial process more task-relevant. By joining the predicted probabilities P and feature maps f_k^l, our domain adversarial model is able to better match and align the distribution of multimodal features in the MCD task. The joint feature h_k^l is computed as follows:

$$h_k^l = \prod \left(f_k^l, P\right) = f \otimes p \tag{2}$$

where p is obtained through a softmax function and the probability map size is guaranteed to be the same as the feature map through adaptive maximum pooling. $\prod()$ is a conditioning strategy implemented by matrix products. For a feature map of size $m \times n$, the objective function of the domain confrontation process is as follows:

$$\begin{aligned}\mathcal{L}_{\text{ADA}} &= \min_{\text{CD}} \max_{\text{DA}} \left(\mathcal{L}_{\text{CD}} - \lambda \cdot \frac{1}{m \times n} \sum_{j=1}^{m \times n} \mathcal{L}_{\text{DA}}\left(D_{\text{dis}}\left(h^l\right), d_j\right)\right) \\ &= \min_{\text{CD}} \max_{\text{DA}} \left(\mathcal{L}_{\text{CD}} - \lambda \cdot \frac{1}{m \times n} \sum_{j=1}^{m \times n} \left[d_j \log p\left(h_j^l\right) \right.\right. \\ &\qquad \left.\left. + \left(1 - d_j\right) \log\left(1 - p\left(h_j^l\right)\right)\right]\right).\end{aligned} \tag{3}$$

The entire $\mathcal{L}_{\text{DA}}$ is calculated based on BCE loss, where d_j represents pseudo-labels of feature source. To enhance the robustness of our adversarial network, an information entropy filter is utilized to screen and filter unstable sample pairs. The uncertainty of the samples is measured by calculating the perceptual weights of the information entropy

$$\omega(x_i) = 1 + e^{\sum_{c=1}^{C} p_{i,c} \log p_{i,c}} \tag{4}$$

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where $p_{i,c}$ is the probability that the ith pixel is predicted to be c class. Based on entropy-aware weights ω, the contribution of the samples in adversarial learning is reset. In summary, the complete domain adversarial loss function is as follows:

$$\mathcal{L}_{\mathrm{ADA}} = \min_{\mathrm{CD}} \max_{\mathrm{DA}} \left(\mathcal{L}_{\mathrm{CD}} - \lambda \cdot \frac{1}{m \times n} \sum_{j=1}^{m \times n} \times \left[\mathbb{E}_{x_1 \sim \mathbb{D}_1} \omega_1 \left(x_j\right) d_j \log p\left(h_j^l\right) + \mathbb{E}_{x_2 \sim \mathbb{D}_2} \omega_2 \left(x_j\right) \left(1 - d_j\right) \log\left(1 - p\left(h_j^l\right)\right) \right] \right). \tag{5}$$

2) Divergence-Aware Contrastive Module: Despite the ability of the feature extractor to stably extract domain-invariant features and the obfuscation of the domain discriminator, there is still a challenge in achieving equilibrium between the adversarial network and the backbone network. This is due to an inherent contradiction between the alignment capability and the detection performance. The performance of the adversarial network can have a significant impact on the difference between the feature distributions of the two modalities. This can result in the fusion operation introducing negative information to the network and affecting the model's detection performance. Conversely, if the adversarial network performs well, the data distributions of the two modalities become strictly aligned. However, the overfitting network can lead to the loss of specific features during the adversarial process. In the MCD task, it is crucial to consider both modality differences and ground-object differences. As described earlier, the MADA helps minimize the negative impact of modal differences on the CD task in the feature space. On the other hand, DCM is specifically designed to enhance feature differences, thereby enhancing the ability to distinguish changing target features. Contrast loss is commonly used in metric learning to learn the distance or similarity of metric tasks. The basic form of contrast loss is as follows:

$$\mathcal{L}_{\mathrm{contras}}(a, p, n) = \frac{1}{2N} \sum_{i=1}^{N} \left[d(a_i, p_i)^2 + \max\left(0, m - d(a_i, n_i)^2\right) \right] \tag{6}$$

which contains an anchor sample a, a positive sample p, and a negative sample n; and d represents the distance between the two samples in the embedding space. The goal of contrastive loss is to reduce the distance between feature representations of samples belonging to the same class while increasing the distance between feature representations of samples belonging to different classes. This approach helps the network learn distinct features and enhance the model generalization capability.

Inspired by this loss, we propose a pixel-level contrast loss to enhance the land cover differences. This loss quantifies the distance between two input features while incorporating change label information during the measurement process. For changed pixels, the feature extraction network is stimulated to learn in the direction of increasing the feature distance between the two points by giving the loss a larger penalty. Conversely, for unchanged pixels, the network keeps the pixel points with small feature distances unchanged but increases the loss values at pixel positions with large feature distances. This process helps the network reduce the distance between unchanged features during optimization. The ultimate goal is to enhance the model's capability to characterize changing features. The calculation is as follows:

$$\begin{aligned} &\mathcal{L}_{\mathrm{DCM}} \\ &= \mathrm{DCM}(D_W, (Y, X_1, X_2)) \\ &= \frac{1}{m \times n} \sum_{i=1}^{m \times n} (Y \cdot [D_W + \eta \cdot \mathbb{I}(D_w \le \alpha)] \\ &\qquad + (1 - Y) \cdot [D_W + \eta \cdot \mathbb{I}(D_w \ge \beta)]). \end{aligned} \tag{7}$$

We set the bi-temporal images X_1 and X_2 and change label Y as a triplet. D_w is the Euclidean distance, which computes the pixel-level difference between the feature maps as follows:

$$D_W(X_1, X_2) = \|X_1 - X_2\|_2 = \sqrt{\sum_{i=1}^{m \times n} \left(x_1^i - x_2^i\right)^2}. \tag{8}$$

The difference in excitation is controlled in DCM using three hyperparameters: η, α, and β. Here, $\mathbb{I}(\cdot)$ L is an indicator function that outputs 1 or 0 depending on the condition, η represents the penalty coefficient, while α and β denote the upper and lower bounds of the distance between two samples, respectively. In the unchanged class, pixel points with a difference distance more than the lower bound β are penalized to reduce the distance within the same feature class. Conversely, in the changed class, pixel points with a difference distance less than the upper bound α are penalized for increasing the distance between different feature classes.

In feature extraction networks, the generality of the features tends to increase with depth. Adversarial tuning of these deep features is effective in reducing distributional differences across different data domains. Hence, the selection of the feature hierarchy significantly affects the performance of adversarial learning. In AFENet, we consider the significance of global features in the CD task. Therefore, we adopt MADA on the underlying multiscale features and regulate the level of adversarial engagement at different scales using learnable parameters. The overall loss of AFENet is computed as follows:

$$\mathcal{L}_{\mathrm{AFE}} = \mathcal{L}_{\mathrm{MADA|DCM}} = \sum_{s-0}^{S} \lambda_s (\mathcal{L}_{\mathrm{ADA}} + \mathcal{L}_{\mathrm{DCM}}). \tag{9}$$

C. MFF for MCD

In the context of the CD task, the algorithm is categorized into three types: early fusion, middle fusion, and late fusion, based on the position of the fusion module. The fusion method used at different stages can impact the model's comprehension of the data. However, there is currently no definitive criterion to determine the superiority or inferiority of different fusion methods. The selection of a fusion method depends on the specific task and characteristics of the data,

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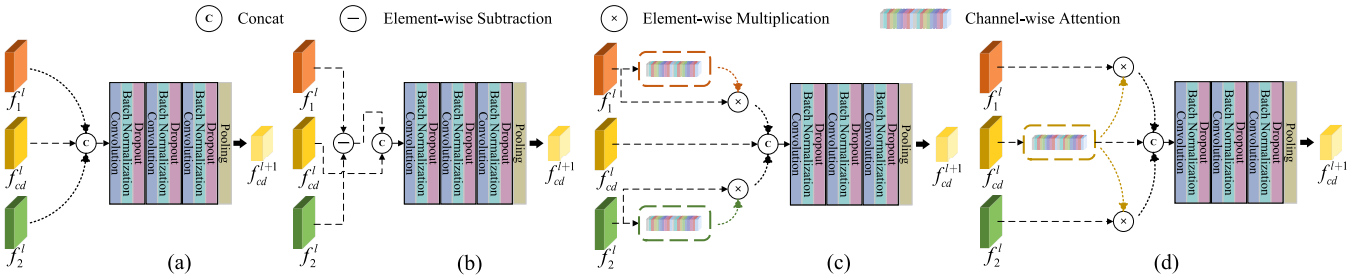


Fig. 3. Structure of the four fusion strategies. (a) Structure of CF. (b) Structure of DF. (c) Structure of AF. (d) Structure of cross AF. In the experimental section, we compare the performance of the four fusion strategies and validate the superiority of our CAF.

Algorithm 1 AFENet for MCD

Input: X_1, X_2: The bi-temporal RS images from two modalities, Y: the CD mask, G_1, G_2: the feature encoder networks, G_{cd}: the feature fusion encoder network, D_{cd}: the decoder network, D_{dis}: the domain discriminator.

while not converged **do**
  Randomly instantiate bi-temporal samples x_1, x_2 and change label y.
  Extract multiscale features f_1^l, f_2^l, f_{cd}^l through G_1, G_2 and G_{cd}.
  for $l = 0$ to $L-1$ **do**
    Fuse and generate deep features by MFF.
  end for
  for $l = L-1$ to 0 **do**
    Retore image from feature map through D_{cd}.
    Skip-connection with encoder feature f_{cd}^l.
  end for
  Generate prediction probability $p_{i,1}$, $p_{i,2}$.
  for $s = 1$ to S **do**
    Generate joint feature h^s by (2).
    Calculate entropy weight ω by (4).
    Calculate $\mathcal{L}_{ADA}^s$ loss by (5).
    Calculate $\mathcal{L}_{DCM}^s$ loss by (7).
  end for
  Calculate CD loss $\mathcal{L}_{CD}$ by (18).
  Calculate AFE loss $\mathcal{L}_{AFE}$ by (9).
  Calculate Final loss $\mathcal{L}_{Final}$ by (19).
  Set parameter gradient to 0.
  Propagate back the loss $\mathcal{L}_{Final}$.
  Update optimizer.
end while

Output: L_{Final}: The Final loss, $\widehat{\theta}_1$, $\widehat{\theta}_2$: parameters for G_1, G_2, $\widehat{\theta}_{cd}$: parameters for G_{cd} and D_{cd}, $\widehat{\theta}_{dis}$: parameters for D_{dis}

which necessitates experimental verification. In this article, we design an MFF framework for information synthesis. The bi-temporal images are fed into two separate CNN streams G_1 and G_2 for feature extraction and feature alignment in the manner described in Section III-B. The aligned features serve as supplementary information to the backbone network stream G_{cd}. They participate in the mid-layer-by-layer fusion of features and skip connection. Additionally, the bi-temporal images are fused at an early stage and fed into the backbone network stream to sense the changing target features. The features from different streams are fused at the same scale and then passed to the following scale. The integrated features are passed through the CD decoder network to generate the final prediction probability. Our backbone network is based on the UNet structure, which retains the symmetric decoder–encoder structure and the skip connection for information transfer. The UNet structure is widely used for its excellent performance in CD tasks.

The research in this section focuses on exploring the feature fusion approach, with a particular emphasis on three key issues: selecting the features for fusion, extracting the features, and performing the fusion. In Section III-B, we introduce the feature equilibrium network and adversarial process as solutions for the first two issues. In this subsection, our focus is on the fusion approach. We present the impact of four different fusion strategies on the mid-term fusion performance, as shown in Fig. 3. Mid-term fusion can enhance the network's ability to perceive both local and global features while preventing premature introduction of unnecessary information. Previous studies [62], [63], [64], [65] have demonstrated the beneficial effects of these features on the MCD task. The features from the three streams, represented as f_1^l, f_2^l, and f_{cd}^l, are fused at the same layer prior to feature extraction using convolutional blocks

$$f_{\text{cd}}^{l+1} = G_{\text{cd}}\left(\mathbb{F}\left(f_{\text{cd}}^l, f_1^l, f_2^l\right)\right) \tag{10}$$

where l is the feature level, $\mathbb{F}$ represents the fusion strategy, and G is the convolutional block used to extract features. The most straightforward strategy is to keep all the multistream features by CF, as shown in Fig. 3(a). The calculation is as follows:

$$\text{CF}\left(f_{\text{cd}}^l, f_1^l, f_2^l\right) = \text{Conv}\left(\text{Concat}\left(f_{\text{cd}}^l, f_1^l, f_2^l\right), 3C_{\text{In}}, C_{\text{Out}}\right). \tag{11}$$

This fusion approach aims to minimize the loss of information but may introduce redundant information and increase the number of computational parameters. In this approach, C_{In} and C_{Out} represent the number of input and output channels of the feature convolution block, and Conv refers to a series of operations in the convolution block, including convolution, batch normalization, and dropout. For the MCD task, the network framework needs to focus on the information about the changes between the bi-temporal images. To achieve this, we propose the second fusion strategy, as illustrated in Fig. 3(b). This strategy, called the difference fusion (DF)

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approach, highlights the difference between two features by computing the difference between the two features after alignment

$$DF\left(f_{\mathrm{cd}}^{l}, f_{1}^{l}, f_{2}^{l}\right) = \mathrm{Conv}\left(\mathrm{Concat}\left(f_{\mathrm{cd}}^{l}, \mathrm{Sub}\left(f_{1}^{l}, f_{2}^{l}\right)\right), 2C_{\mathrm{In}}, C_{\mathrm{Out}}\right) \tag{12}$$

where Sub is an element-level subtraction strategy that helps the model understand different parts of the features and reduces the number of parameters. However, it may result in the loss of valid information, especially when there is a complementary relationship between f_1^l and f_2^l. The attention mechanism plays an important role in artificial neural networks and has been extensively studied. In feature fusion, the attention mechanism allows for adaptive adjustment of the importance of different features, thereby enhancing the model's focus on key information. Building on previous research, we aim to integrate information from different modalities using the attention mechanism effectively. By dynamically adjusting the weight of each modality, the model can better leverage the advantages of features from different sources and improve the expressive ability of the fused features. Therefore, in the attention fusion (AF) strategy, we introduce the channel attention mechanism (CAM) separately in the f_1^l and f_2^l feature streams. As shown in Fig. 3(c), each feature undergoes individual attention before being fused with f_{cd}^l. This process can be represented by the following equation:

$$AF\left(f_{\mathrm{cd}}^{l}, f_{1}^{l}, f_{2}^{l}\right) = \mathrm{Conv}\left(\mathrm{Concat}\left(f_{\mathrm{cd}}^{l}, \mathrm{CAM}\left(f_{1}^{l}\right), \mathrm{CAM}\left(f_{2}^{l}\right)\right), 3C_{\mathrm{In}}, C_{\mathrm{Out}}\right). \tag{13}$$

For the input feature $f_k^l \in \mathbb{R}^{C\times H\times W}$, where C is the number of channels, and H and W are the height and width of the feature map, respectively. The CAM is computed as follows:

$$\mathrm{CAM}\left(f_{k}^{l}\right) = f_{k}^{l} \cdot \omega_{\mathrm{CAM}}\left(f_{k}^{l}\right) \tag{14}$$

where ω_{CAM} is the attentional weight of each channel. CAM generally uses global average pooling (GAP) to obtain global information. The weights are then generated through a fully connected layer $F(\cdot)$ and applied to the features. The attention weights are generated as follows:

$$\omega_{\mathrm{CAM}}\left(f_{k}^{l}\right) = \frac{FC\left(f_{k}^{l}\right)_{c}}{\sum_{i}^{C} FC\left(f_{k}^{l}\right)_{i}} \tag{15}$$

$$FC\left(f_{k}^{l}\right)_{c} = \sigma\left(\omega_{2}\delta\left(\omega_{1}\mathrm{GAP}\left(f_{k}^{l}\right)_{c} + b_{1}\right) + b_{2}\right) \tag{16}$$

where δ is the activation function, σ is the sigmoid function mapping the weights to $[-1,1]$. The learnable parameters, ω_1, b_1, ω_2, and b_2, are also incorporated in the process. The AF strategy is designed to adaptively adjust the contribution of each feature channel during the fusion process. This capability enables the feature extraction network to effectively capture small-scale target features and enhance its sensitivity to subtle changes. However, the fusion between different streams is still relatively independent since the attention mechanism is applied individually in the two feature streams. To enhance the interaction between multistream features and achieve a more task-friendly fusion method, we propose the CAF strategy. This strategy primarily considers the characteristics of the backbone network when fusing multistream features. Since the backbone network focuses on perceiving changing targets, we aim to align the areas of interest in the two streams with those in the backbone network. Fig. 3(d) illustrates the framework of CAF. The process can be expressed as follows:

$$\mathrm{CAF}\left(f_{\mathrm{cd}}^{l}, f_{1}^{l}, f_{2}^{l}\right) = \mathrm{Conv}\left(\mathrm{Concat}\left(f_{\mathrm{cd}}^{l}, f_{1}^{l} \cdot \omega_{\mathrm{CAM}}\left(f_{\mathrm{cd}}^{l}\right), f_{2}^{l} \cdot \omega_{\mathrm{CAM}}\left(f_{\mathrm{cd}}^{l}\right)\right), 3C_{\mathrm{In}}, C_{\mathrm{Out}}\right). \tag{17}$$

This strategy adds the CAM to the f_{cd}^l stream while applying the generated channel weights ω_{CAM} to the f_1^l and f_2^l feature streams, respectively. The goal is to ensure that all the multistream features concentrate on the same region of interest for change. The features integrated by the strategies mentioned above are inputted into the CD decoder D_{cd}, which generates the change probability map. Our research primarily investigates the influence of multistream feature interaction and fusion strategies on the MCD task. Therefore, we employ a simple encoder–decoder structure in this article. With the classification probability $p_{i,c}$, we can calculate the loss using the cross-entropy (CE) function

$$\mathcal{L}_{\mathrm{CD}} = \mathrm{CE}(X, Y) = \frac{1}{M \times N} \sum_{i=1}^{M\times N} \sum_{c=1}^{C} y_{i,c} \cdot \log\left(p_{i,c}\right). \tag{18}$$

The reason why we use CE instead of BCE is that we need two types of probabilities, changed and unchanged, to participate in the ADA process in Section III-B. In conclusion, our overall loss is determined by the CD task loss and the AFE loss. The feature extraction network is trained to achieve an equilibrium state by considering the AFE loss, which helps improve the adaptability of the extracted features for the MCD task. We will provide experimental evidence to support our claim in the upcoming section

$$\mathcal{L}_{\mathrm{Final}} = \mathcal{L}_{\mathrm{CD}} + \mathcal{L}_{\mathrm{AFE}}. \tag{19}$$

IV. Experiment and Result

In this section, we introduce our experimental datasets and provide details on their implementation. To evaluate the effectiveness of our method, we conducted a series of ablation experiments. Furthermore, we demonstrated the advanced nature of our method by comparing it with existing methods.

A. Datasets Description

MT-HTCD is a satellite-UAV multimodal dataset proposed by Shao et al. [66], which captures Chisinau and its surrounding area in Moldova. The satellite image was taken in 2008 and has a size of 11K × 15K pixels in 59.71 cm ground resolution. The UAV image has 1.38M × 1.04M pixel size and 7.47 cm ground resolution, which was shot in 2020 and consists of 15 blocks. The annotation of changed regions mainly focuses on the buildings, roads, and other urban features.

MT-Wuhan is an optical-SAR heterogeneous dataset for CD tasks, which includes an optical image and an HH-polarized

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SAR image over Wuhan. The dual-temporal images were acquired by Sentinel-2 in 2017 and COSMO-SkyMed in 2020. The ground resolution of them is 10 and 3 m, respectively. Pre-processing, such as geometric correction, transformation, and interpolation, are applied to unify the size and balance the feature distribution of SAR and optical images. The final size of change labels and images is 256 × 256 pixels, and the number of unchanged pixels is about five times the changed. This dataset is collated by Zhang et al. [54].

B. Implement Details

Our model is implemented based on the PyTorch framework and trained on an Nvidia RTX 3090 GPU. During the training process, we update the parameters using an Adam optimizer with an initial learning rate of 0.001. We also employ a multistep learning rate scheduler, where the learning rate is adjusted at specified milestones with a decay factor of 0.8. We set the batch size to 4 to accommodate GPU memory constraints and monitor the model's learning progress over 1000 epochs.

In the MT-Wuhan dataset, we maintain the original author's division, comprising 522 pairs of training samples and 112 pairs of testing samples. The bi-temporal images and labels are all of size 256 × 256, which can be directly read and inputted into the network for learning. As for the MT-HTCD dataset, the authors do not provide the divided version, so we divide the 3772 pairs of images in the 7:2:1 ratio. Seventy percent of the data is used for training and ten percent for testing. The satellite image is 256 × 256, the aerial image is 2047 × 2047, and the label size is 1528 × 1528. We standardize the image and label sizes to 256 × 256 before feeding into the network. Our experiment does not utilize strategies such as pre-trained models, data augmentation, etc.

1) Network Details: In accordance with the framework depicted in Fig. 2, our model is composed of several networks: two feature extraction networks, a feature fusion network, a task head network, and a discriminator network. The feature extraction network is comprised of a five-layer CNN module that includes convolution, pooling, and dropout layers. The number of channels gradually increases at each scale (32-64-128-256-256). At each scale, three sets of features of the same size are inputted into the feature fusion network for fusion. Deeper features are then extracted by utilizing two convolutional modules (64-128-256-512-512). The network structure for the feature fusion network and the task head network is based on an encoder–decoder structure similar to UNet. The decoder network recovers features layer-by-layer (512-256-128-64-32) using the transposed convolution operations while preserving local–global information through skip connections. The discriminator network predicts the probability map of feature sources with the same number of channels but different sizes at different scales. For different probability map sizes, corresponding domain pseudo-labels are generated to compute the adversarial loss. During the adversarial training process, to guide the generator to better adapt to the target domain, we set a dynamically varying weight coefficient ω_{coeff} to adjust the gradient inversion's strength. The calculation is as follows:

$$\omega_{\text{coeff}} = 2 \times \left(\frac{\text{up-bottom}}{1 + e^{-\sigma \cdot \text{iter}/\,\text{max_iter}}} - (\text{up-bottom}) + \text{bottom} \right) \tag{20}$$

where iter is the current iteration number and σ is the function slope. In our experiments, we set σ, up, and bottom to 10, 0, and 1, respectively. This weight coefficient enables the strength of the gradient inversion to gradually increase with the number of training iterations.

TABLE I
QUANTITATIVE RESULT (IN %) OF COMPARISON METHODS ON MT-WUHAN DATASET

Method	Evaluation Criterion				
	OA	F1	Recall	Precision	mIOU
FC-EF(2018)	87.79	55.39	54.21	56.63	62.55
FC-Siam-Conc(2018)	87.92	54.39	51.52	57.61	62.17
FC-Siam-Diff(2018)	87.45	52.00	48.59	55.93	60.84
STANet(2020)	82.99	47.01	53.96	41.64	56.16
SNUNet(2021)	**95.96**	44.39	46.18	42.74	62.21
SUNet(2021)	45.64	28.69	**78.19**	17.57	27.86
BIT(2021)	83.89	45.74	48.55	43.24	56.19
LGPNet(2021)	88.25	46.41	46.60	46.21	55.24
MSCDNet(2022)	88.40	54.43	49.53	60.40	62.46
MSDC(2023)	87.49	54.49	53.56	55.46	61.96
AERNet(2023)	85.36	50.84	54.12	47.93	59.12
HAFF(2023)	85.18	46.96	46.92	46.99	57.41
STADE(2024)	84.19	45.55	47.28	43.94	56.28
Ours	89.12	**60.03**	58.43	**61.72**	**65.52**

2) Comparison Methods: We compared the proposed AFENet with several existing CD methods, including FC-EF [23], FC-Siam-Conc [23], FC-Siam-Diff [23], STANet [35], SNUNet [25], SUNet [66], BIT [29], LGPNet [67], MSCDNet [54], MSDC [68], AERNet [69], HAFF [70], and STADE [71]. The first three fully convolution-based frameworks are mainly classical approaches in CD task. STANet utilizes two different attention mechanisms to focus on spatio-temporal information. SNUNet incorporates a tighter feature connection by introducing the attention mechanism. SUNet introduces edge-assisted information for heterogeneous RS images. BIT is the first CD method to utilize the transformer operation. LGPNet effectively integrates local details and global features. MSCDNet focuses on the MCD task and introduces the maximum mean discrepancy (MMD) in DA to address the differences in heterogeneous images. In MSDC, the discrete cosine transform (DCT) is utilized to extract high-frequency information and enhance the representation of changing features by incorporating frequency-domain information. AERNet incorporates an edge refinement module to improve the model's capability to recognize the details of changing edges. HAFF enhances the global semantic features from multiperspective ways through a novel hierarchical AF network. STADE addresses the challenges of class imbalance and pseudo-change through a CD difference-enhancement module (CDDM) and a temporal memory module (TMM). Most comparison methods have excellent performance on homogeneous data. We also choose the networks tailored

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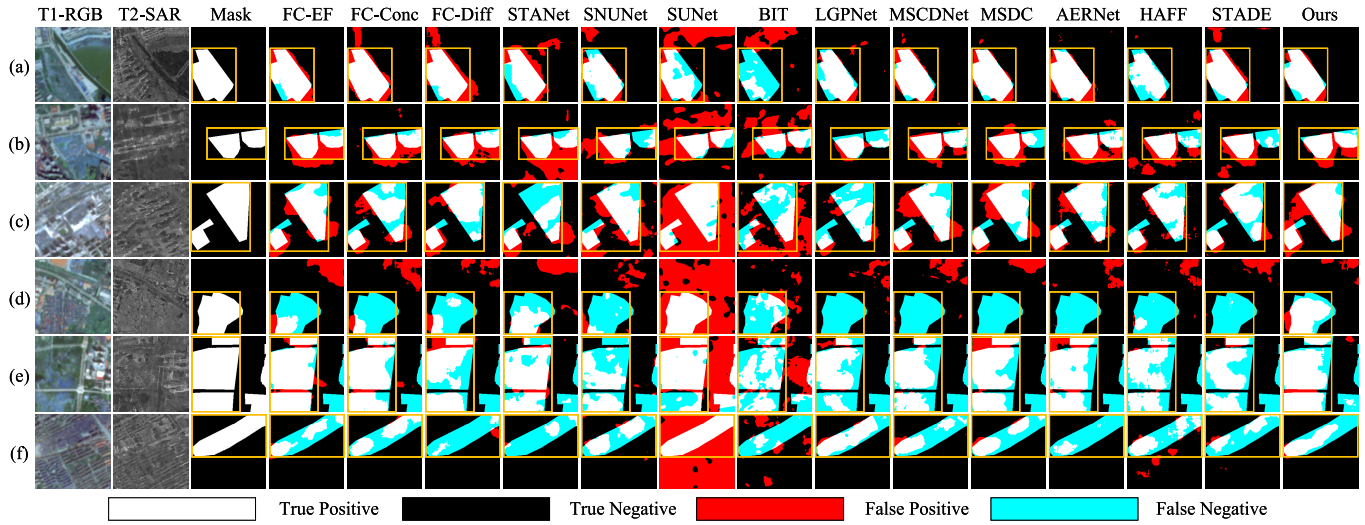


Fig. 4. Visualization of the comparison results on MT-Wuhan dataset. Rows (a)–(f) represent the visualization results of six different samples. The change maps are colored for better observation. Additionally, yellow boxes are used to highlight the change details.

for heterogeneous data, such as SUNet, MSCDNet, and HAFF, to demonstrate the effectiveness of our approach comprehensively. To ensure the reliability of the comparison, all experiments were performed in the same configuration environment. We made an effort to use the source code and default parameter settings published by the owners for each comparison method. For methods that are not open source, we reproduce the model design strictly as described in the original article.

3) Evaluation Metrics: In order to assess the model performance, we calculate various evaluation metrics using true positive (TP), true negative (TN), false positive (FP), and false negative (FN) from the statistical confusion matrix. In experiments, we employ five commonly used evaluation metrics to validate the detection performance of different methods. Precision measures the accuracy of the TP samples' predictions; recall indicates the proportion of correctly predicted TP samples by the model, and the $F1$ score represents the average result of reconciling precision and recall. Additionally, mIOU is a widely used evaluation metric in semantic segmentation tasks, which measures the segmentation accuracy of the model for each category. Lastly, OA denotes the model's classification accuracy across all categories. The formulas for these metrics are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}.$$

C. Experimental Results

1) Experimental Results on MT-Wuhan Dataset: The proposed method was first validated on the MT-Wuhan dataset, and the comparison results are presented in Table I, where bolded values represent the best results and underlined values are the second-best results. Due to the heterogeneous nature and significant modal differences of the bi-temporal images in MT-Wuhan, the existing methods generally yield poor detection results. Among the compared methods, only six achieved $F1$ scores above 50%, with SUNet scoring only 28.69%. This can be attributed to the fact that SUNet is designed for heterogeneous data of different sizes and incorporates edge information during feature extraction, necessitating high-quality edge image extraction. The edge information extracted in the SAR scenario is often cluttered, which leads to the lowest precision of SUNet in our experiment. It is worth noting that the highest recall corresponds to the lowest $F1$ score for SUNet, indicating that SUNet tends to find most change regions but a lot of false positives exist. The FC-EF, as a simple encoder–decoder structure, achieves the second-best $F1$ score and mIOU by simply concatenating bi-temporal images into the network and extracting change features directly. In contrast to other methods with various modules, we can learn that the stacked module approach tends to negatively affect the detection performance when dealing with multimodal data with large differences. By comparing the detection results of the methods tailored for heterogeneous data, we can know that these methods, SUNet, MSCDNet, and HAFF, are highly specific, and their detection performance fluctuates greatly on different data. Our proposed AFENet achieves state-of-the-art (SOTA) performance in $F1$ score, precision, and mIOU. The OA and recall rank second-best in comparison methods. It reflects the fact that our method can distinguish between changed and unchanged regions relatively accurately, but still miss some changed regions. Besides, category imbalance is a common problem in the CD task, where unchanged regions tend to occupy a larger proportion. When the model is weak in detecting unchanged regions, it also leads to lower OA. Therefore, the $F1$ score and mIOU are commonly used in CD tasks to measure the overall performance of the model.

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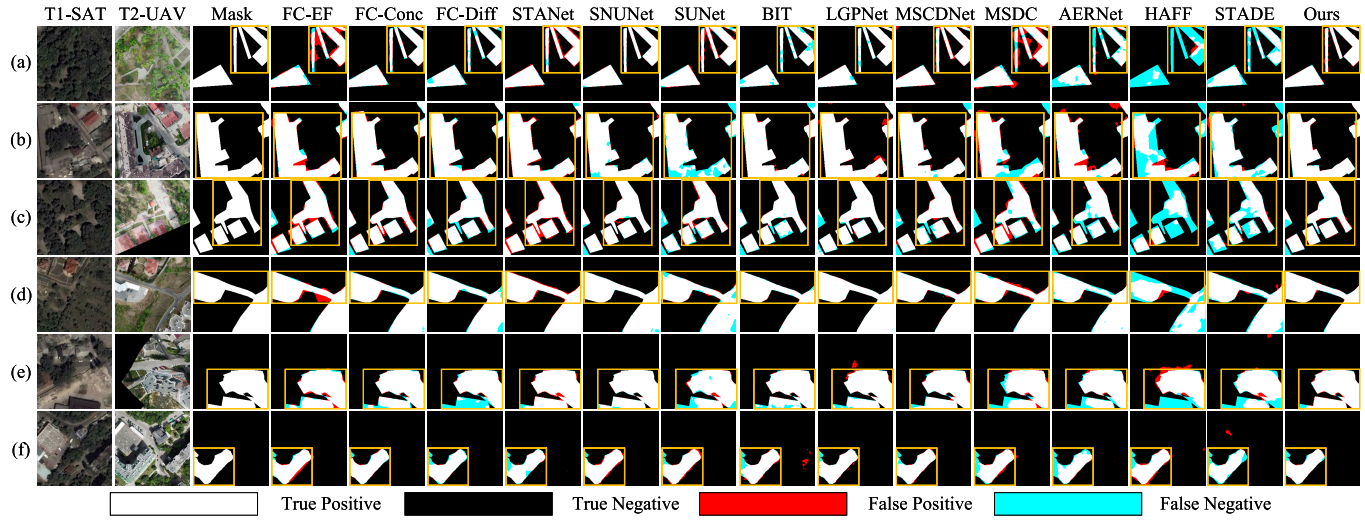


Fig. 5. Visualization of the comparison results on MT-HTCD dataset. Rows (a)–(f) represent the visualization results of six different samples. The change maps are colored for better observation. Additionally, yellow boxes are used to highlight the change details.

In addition, the detection results of six pairs of samples have been visualized in Fig. 4. We can see significant disparities between the feature representations of RGB images and SAR images. Some regions that have changed are hard to detect even through visual observation, which poses a challenge for the network model. To facilitate better observation of performance differences, we have stained the visualization results with four colors, representing the TP, TN, FP, and FN in the confusion matrix. The visualization results in Fig. 4 demonstrate the superiority of our method. There is no wide range of missed detections, such as in LGPNet (d), MSCDNet (d), MSDC (d), AERNet (d), and STADE (d). Our approach best preserves the internal continuity of the change region, avoiding localized missed detection within the change region as observed in STANet (e), SNUNet (e), BIT (e), and HAFF (e). In comparison to existing methods, our model observed superior prediction results, but there are still noticeable shortcomings in detecting change edges and avoiding false detection for the unchanged region. As illustrated in samples (b), (c), (e), and (f), there are more missed or false detections that tend to occur at the edges of our detection results.

2) Experimental Results on MT-HTCD Dataset: The original authors defined the MT-HTCD dataset as a heterogeneous RS dataset, including bi-temporal images acquired from satellite and UAV. Despite the differences in acquisition sensors, the modal differences after imaging are not as significant as in MT-Wuhan. The detection performance on HTCD data can be evidence of the method's generalizability in this article rather than being limited to specific datasets. Table II presents the evaluation results, where bolded values are the best results and underlined values are the second-best results. Due to the highly unbalanced positive and negative samples in MT-HTCD data, a significant number of unchanged pixels were correctly recognized, leading to high OA results of over 99%. Despite OA, our AFENet demonstrates superior performance in $F1$, recall, precision, and mIOU. The performance of SNUNet is very close to our method. SNUNet achieves an $F1$ score of only 1.1% lower than our method, 1.02% lower in recall, and 1.08% lower in mIOU. This can be attributed to the tight feature extraction structure of SNUNet and the hybrid attentional mechanism of ECAM. The ECAM mechanism effectively learns the changing target features in supervised learning with a relatively large amount of data, resulting in good detection performance. The second-best precision is achieved by LGPNet, which utilizes a series of attention mechanisms and effectively integrates local details and global features through pyramid feature extraction. Among the comparison methods tailored for heterogeneous data, HAFF achieves the worst performance, which may be caused by its smallest number of model parameters. Such small networks cannot learn feature representations from large-scale data well. MSCDNet performs well on both multimodal datasets, thanks to its domain-adaptive feature constraint module that aligns modal differences effectively. Besides, the detection performance of FC-Siam-Conc is close to the MSCDNet. We know from the

TABLE II
QUANTITATIVE RESULT (IN %) OF COMPARISON METHODS ON MT-HTCD DATASET

Method	Evaluation Criterion				
	OA	F1	Recall	Precision	mIOU
FC-EF(2018)	99.47	90.20	89.44	90.98	90.81
FC-Siam-Conc(2018)	99.71	94.49	92.98	96.04	94.63
FC-Siam-Diff(2018)	99.64	92.98	88.97	97.37	93.26
STANet(2020)	99.54	91.02	92.42	89.65	91.52
SNUNet(2021)	<u>99.96</u>	<u>96.05</u>	<u>94.42</u>	97.43	<u>96.18</u>
SUNet(2021)	99.54	90.60	87.11	94.39	91.18
BIT(2021)	99.89	92.65	88.90	96.72	93.09
LGPNet(2021)	99.73	95.63	93.42	<u>97.94</u>	95.67
MSCDNet(2022)	99.71	94.17	91.44	97.07	94.34
MSDC(2023)	99.23	85.82	82.37	89.58	87.21
AERNet(2023)	99.26	84.58	79.79	89.98	86.24
HAFF(2023)	98.17	60.02	51.38	72.15	70.51
STADE(2024)	98.85	72.39	71.61	73.15	77.77
Ours	**99.97**	**97.15**	**95.71**	**98.64**	**97.26**

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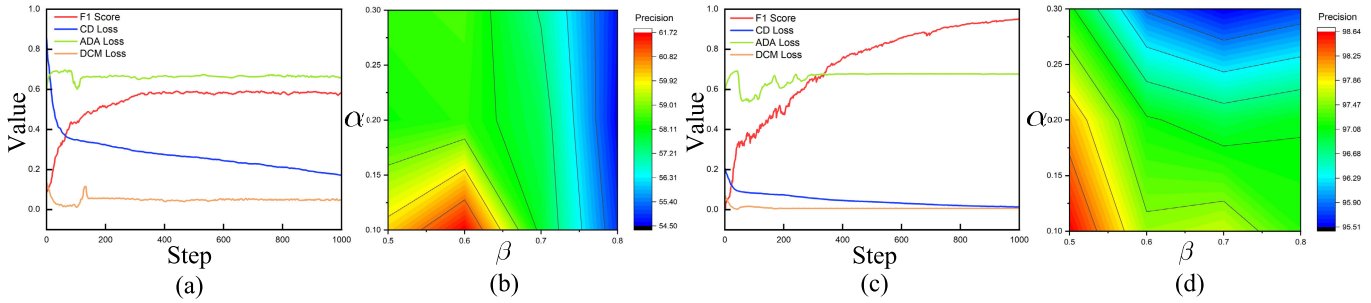


Fig. 6. Visualization of the AFENet training process and hyperparameter analysis. (a) Variation of loss functions and $F1$ score on MT-Wuhan dataset. (b) Accuracy effect of parameter settings on MT-Wuhan dataset. (c) Variation of loss functions and $F1$ score on MT-HTCD dataset. (d) Accuracy effect of parameter settings on MT-HTCD dataset.

TABLE III
ABLATION STUDIES (IN %) FOR DIFFERENT COMBINATIONS OF LOSS TERMS

Method	MT-Wuhan (sar/rgb)					MT-HTCD (sat/uav)				
	OA	F1	Recall	Precision	mIOU	OA	F1	Recall	Precision	mIOU
$\mathcal{L}_{CD}$	88.01	49.06	41.28	60.46	59.89	99.15	84.47	86.33	82.68	86.12
$\mathcal{L}_{CD}$+$\mathcal{L}_{ADA}$	**89.48**	56.43	48.72	**67.05**	64.01	99.86	94.12	90.58	97.94	94.29
$\mathcal{L}_{CD}$+$\mathcal{L}_{ADA}$+$\mathcal{L}_{DCM}$	89.12	**60.03**	**58.43**	61.72	**65.52**	**99.97**	**97.15**	**95.71**	**98.64**	**97.26**

experimental results that supervised methods generally achieve great detection performance when the quality of the data is high enough. Most of the comparison methods can achieve $F1$ scores above 90%.

The visualization results of the comparison methods are shown in Fig. 5. The figure shows that the bi-temporal images in MT-HTCD have fewer modal differences and higher image resolution than MT-Wuhan. As a result, comparison methods can detect most change regions correctly and the overlap between prediction results and masks is high. We can only observe performance differences between methods in specific samples. For instance, in sample (a), there is dense vegetation on both sides of the road where the change occurs, leading to the change region being obscured partially. As shown in the visualization, few false detections and missed detections exist around roads. Another complex case is shown in sample (c); there are multiple change regions with only small gaps, which can easily lead to the model recognizing these regions as a whole and making it difficult to distinguish the change boundaries. It can be perceived in the FC-EF (c), STANet (c), and SUNet (c). However, the visualization results of our AFENet present the fewest false positives and false negatives and the highest similarity to change masks.

D. Ablation Analysis

In addition to detecting results, it is necessary to validate the effectiveness of each module in the network and analyze the hyperparameter settings in these modules. Our model mainly comprises two parts: AFENet is utilized to guide feature extraction and align the features, while the aligned features are integrated by MFF and predicted for changed targets. In this section, we investigate the effects of AFENet and MFF on MCD performance through ablation experiments. Furthermore, we validate the effectiveness of the MADA and DCM in AFENet separately.

1) Effectiveness of AFENet: Our AFENet is composed of MADA and DCM. MADA utilizes a similar adversarial framework as CDAN to drive the feature extractor in extracting domain-invariant features between two modalities. This is achieved by confusing the discriminator, resulting in feature alignment in the feature space. However, unlike CDAN, we perform the adaptation at multiple scales and introduce a regularization term DCM in the adversarial process. AFENet addresses the challenge of maintaining equilibrium between modal and feature differences in adversarial learning, thereby making network optimization more conducive to the CD task. The detection outcomes obtained by employing various combinations of loss functions during model training are presented in Table III. To eliminate the influence of other variables, we conducted three sets of comparative experiments with consistent fusion strategy and training parameters. These experiments were validated on two datasets, respectively. The first set of experiments directly fuses the multimodal features and then detects the changes. The MADA module was added to fuse the aligned features before detection in the second set of experiments. The third set of experiments involved the complete AFENet with DCM added to MADA, followed by feature fusion and CD. The results in Table III show that without adversarial learning, the model's detection performance is significantly lower compared to the other two groups, with the $F1$ score dropping by 12.68% on the MT-HTCD. This is primarily because unprocessed multimodal features fail to serve as complementary information to the MCD network and introduce noisy information, thereby reducing the model's detection performance. The experimental results show a significant improvement in the $F1$ score with the addition of MADA. In the last set of experiments, our complete AFENet outperforms other models in all terms of evaluation metrics on MT-HTCD data. In the MT-Wuhan dataset, although precision decreases, there is a noticeable improvement in recall. This

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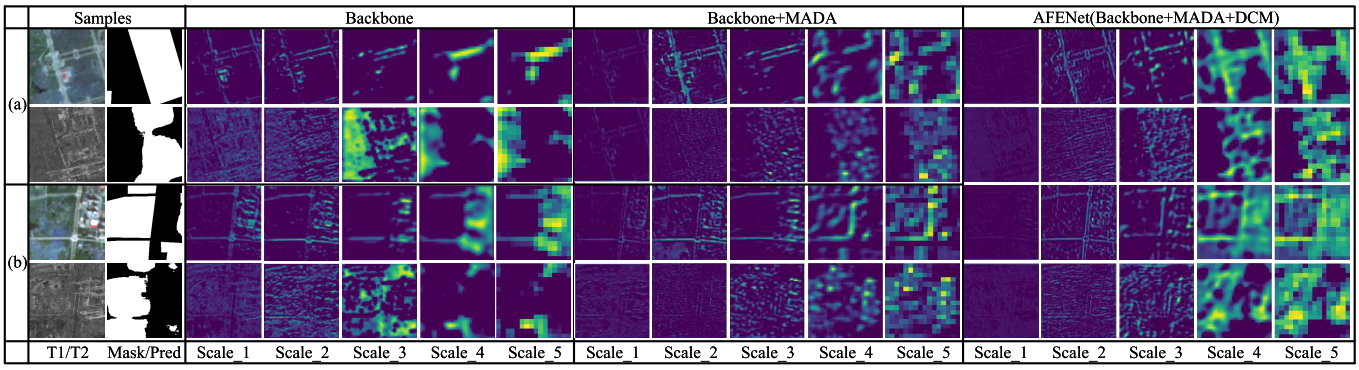


Fig. 7. Visualization of feature maps on the MT-Wuhan dataset. In each row, we visualize the feature maps extracted from two branch networks. The images in different columns represent the extracted features from the well-training networks with different components. Rows (a)–(b) represent the visualization results of two pairs of bi-temporal images.

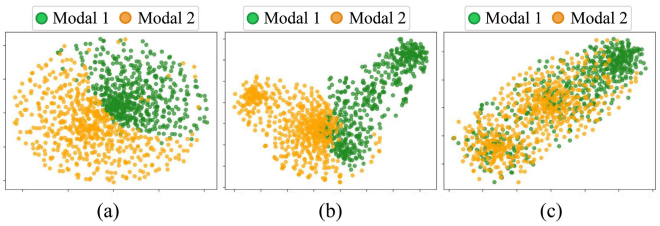


Fig. 8. Visualization of feature representations by t-SNE on MT-Wuhan dataset. (a) Visualization of feature maps in initial stage. (b) Visualization of feature maps in middle stage. (c) Visualization of feature maps in well-trained stage.

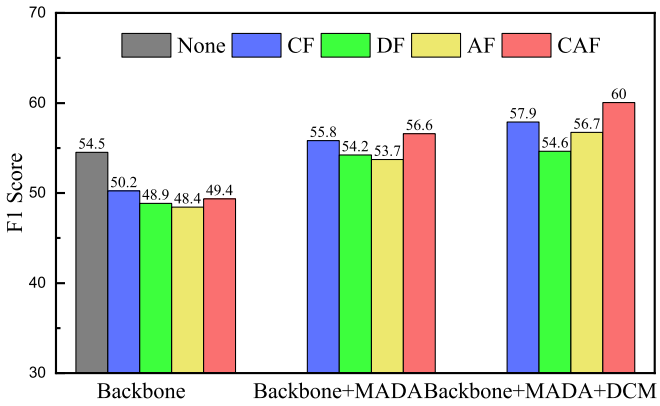


Fig. 9. Comparison result of $F1$ score with different fusion strategies on MT-Wuhan dataset.

result aligns with the motivation of our DCM, which aims to prevent the loss of changed target features during the strict alignment process. By introducing DCM, our model achieves a more balanced detection performance in the MCD task. Overall, our method achieves SOTA results in both comprehensive performance evaluation metrics, including $F1$ score and mIOU.

In the DCM, two hyperparameters, α and β, are set as the upper and lower bounds of the tolerance threshold for feature differences in the comparison module, respectively. These parameters control the sensitivity of the network to unchanged and changed pixels by adjusting the upper and lower bounds of the feature distance, thereby enhancing the discriminative power of the model. For the experiment, we selected 3×4 sets of parameters for the settings of α and β. The precision results under different parameter settings are depicted in Fig. 6(b) and (d). It can be observed that the model achieves the highest detection accuracy for the MT-Wuhan dataset when $\alpha = 0.6$ and $\beta = 0.1$, while the MT-HTCD data exhibits the highest precision result when $\alpha = 0.5$ and $\beta = 0.1$. Visualizing the training process provides a better understanding of the roles played by MADA and DCM in AFENet. In Fig. 6(a) and (c), the four different colored curves represent the loss of the CD task, the loss of MADA, the loss of DCM, and the change of $F1$ score, respectively. It is observed that both MADA and DCM converge as the number of training steps increases. The CD loss also decreases during this process, resulting in a stable $F1$ score. The implementation of feature equilibrium can be more clearly observed in the training process of MT-Wuhan data. Initially, the network focuses on modal differences, but, as training progresses, the differences between the two groups of features diminish. DCM influences the network at approximately 100 steps, and the feature difference stabilizes after a slight increase. The final post-stabilization difference loss is slightly higher than the loss achieved in the first phase. We conclude that the network achieves equilibrium for multimodal features in this state.

We conducted a visualization of the features extracted by AFENet at multiple scales, both before and after adversarial learning. In Fig. 7, it is evident that the region of interest in the feature map demonstrates a perception of changing regions following adversarial learning. Furthermore, the regions of interest in the two modal feature maps tend to coincide. We further performed a visualization of the distribution of deep features using t-SNE. Fig. 8 reveals that the feature distributions of the two modalities become more similar under the influence of AFENet, thereby addressing the interference caused by modal differences to some extent.

2) Effectiveness of MFF: The features extracted by AFENet were integrated by MFF and used to generate change probability maps. We conducted the comparative experiments of the four fusion strategies shown in Fig. 3 and demonstrated the impact caused by different fusion strategies on the $F1$ score through a bar graph. Based on the results presented in Fig. 9, it can be observed that the detection performance without feature fusion is superior to all four fusion strategies when MADA and DCM are not added. However, when MADA and DCM are added, different fusion strategies exhibit varying

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TABLE IV
STATISTICS ON PARAMETERS AND COMPUTATIONAL REQUIREMENTS OF THE COMPARISON METHODS

	FC-EF	FC-Siam-Conc	FC-Siam-Diff	STANet	SNUNet	SUNet	BIT	LGPNet	MSCDNet	MSDC	AERNet	HAFF	STADE	Ours
Params(M)	21.55	24.68	21.55	16.93	12.04	15.56	3.49	70.88	45.81	55.39	25.36	2.37	10.49	39.70
Macs(G)	223.78	335.19	296.53	52.65	219.29	159.53	42.46	237.01	176.43	236.38	51.27	118.88	47.92	398.58

performance. The $F1$ score results indicate that the CAF strategy performs the best in a state of feature equilibrium. The experimental results shown earlier and the ablation experiments are also based on the results obtained from the CAF strategy.

3) Statistics of Model Computation: Finally, we analyzed the computational complexity of the proposed model by counting the number of parameters and multiply accumulate operations (MACs). MACs measure of the computational demand of the model, with higher MACs indicating greater computational complexity. To determine the size and complexity of the model, we generated random samples with a batch size of 4. The statistical results are presented in Table IV, which demonstrate that our network includes multiple independent feature extraction networks and a domain discriminator, resulting in a higher number of MACs compared to other methods. Therefore, the computational demand for our model is greater. In the comparison method, our AFENet is medium-sized, with an overall parameter count of 39.7 MB.

V. CONCLUSION AND FUTURE WORKS

In this article, we propose an end-to-end MCD method called the AFENet, which addresses the issue of modal differences between multimodal images, especially in heterogeneous situations. We establish an additional adversarial optimization in AFENet to address the equilibrium problem between modal differences and land cover changes. AFENet achieves the feature alignment to reduce the modal difference through MADA. Meanwhile, a DCM is designed to enhance the land cover change by adjusting the mutual information between changed and unchanged pixels. Through the joint effect of MADA and DCM, the encoder network can extract domain-aligned features while maximizing the discriminability of the change target. The extracted features will be integrated by a CAF strategy in MFF and utilized to generate change maps. We evaluated the detection performance of AFENet on two heterogeneous datasets and validated the effectiveness of MADA, DCM, and CAF through a series of ablation experiments. The experimental results show that our approach achieves the SOTA performance on the $F1$ score and mIOU metrics, which can demonstrate the overall superiority of our AFENet.

Compared to the existing methods, our AFENet solves the negative impact of modal differences through a novel adversarial equilibrium approach and effectively improves detection accuracy. However, the performance of AFENet highly relies on the quality of the label information. Besides, the complex network components and the pixel-level supervised training lead to a mass of parameters, which can increase the training cost and time consumption of the network. In our future work, we will further explore weakly supervised and unsupervised approaches. We aim to expand the application scenarios and improve the generalization ability of MCD methods.

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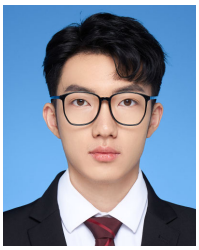
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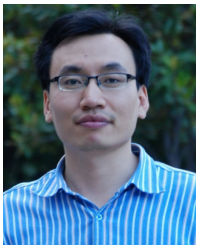
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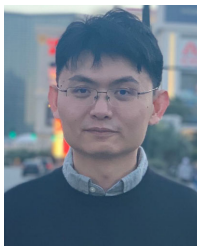
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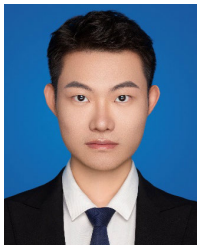
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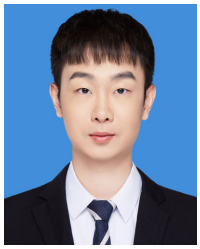
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